

EA-0029 - AQELYN Engineering Archive

AQELYN - EA-0029 Engineering Archive

IS-029 - AQELYN SaaS Security Posture Management (SSPM) Intelligence Engine

Archive ID: EA-0029

Implementation Specification: IS-029

Component: AQELYN SaaS Security Posture Management (SSPM) Intelligence Engine

Project: AQELYN

System Type: Cyber Security Operating Environment

Status: COMPLETE

Repository Impact: No top-level repository structure changes

Breaking Changes: None

Predecessor Archives: EA-0001 through EA-0028

Next Specification: IS-030 - AQELYN Software Supply Chain Security & SBOM Intelligence Engine

Document Control

| Field | Value |

---|---

| Document | Engineering Archive EA-0029 |

| Specification | IS-029 - AQELYN SaaS Security Posture Management (SSPM) Intelligence Engine |

| Publication Format | Markdown, PDF, HTML, ZIP |

| Source of Truth | MD/EA-0029.md |

| Archive Rule | Implementation Specification -> Engineering Archive -> Continue |

| Repository Rule | Fixed repository structure; no redesign |

| Completion State | IS-029 complete; EA-0029 generated |

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment.

Completed engineering archive chain:

| Engineering Archive | Implementation Specification | Component |

---|---|---

| EA-0001 | IS-001 | AQELYN Kernel |

| EA-0002 | IS-002 | Universal Object Model |

| EA-0003 | IS-003 | AQELYN Event Bus |

| EA-0004 | IS-004 | AQELYN Evidence Engine |

- | EA-0005 | IS-005 | AQELYN Knowledge Graph |
- | EA-0006 | IS-006 | AQELYN Trust Engine |
- | EA-0007 | IS-007 | AQELYN Mission Engine |
- | EA-0008 | IS-008 | AQELYN Workflow Engine |
- | EA-0009 | IS-009 | AQELYN Policy Engine |
- | EA-0010 | IS-010 | AQELYN Compliance & Governance Engine |
- | EA-0011 | IS-011 | AQELYN Identity & Access Governance Engine |
- | EA-0012 | IS-012 | AQELYN Asset & Configuration Governance Engine |
- | EA-0013 | IS-013 | AQELYN Risk Intelligence Engine |
- | EA-0014 | IS-014 | AQELYN Threat Intelligence Fusion Engine |
- | EA-0015 | IS-015 | AQELYN Security Operations Engine |
- | EA-0016 | IS-016 | AQELYN Digital Forensics Engine |
- | EA-0017 | IS-017 | AQELYN Threat Detection & Analytics Engine |
- | EA-0018 | IS-018 | AQELYN Automated Response & Orchestration Engine |
- | EA-0019 | IS-019 | AQELYN Security Data Lake & Telemetry Platform |
- | EA-0020 | IS-020 | AQELYN AI Decision Intelligence Engine |
- | EA-0021 | IS-021 | AQELYN Predictive Analytics & Forecasting Engine |
- | EA-0022 | IS-022 | AQELYN Executive Intelligence & Strategic Reporting Engine |
- | EA-0023 | IS-023 | AQELYN Threat Exposure & Attack Surface Management Engine |
- | EA-0024 | IS-024 | AQELYN Vulnerability Intelligence & Prioritization Engine |
- | EA-0025 | IS-025 | AQELYN Cyber Asset Discovery & Inventory Intelligence Engine |
- | EA-0026 | IS-026 | AQELYN Configuration Compliance & Drift Intelligence Engine |
- | EA-0027 | IS-027 | AQELYN Identity Threat Detection & Behavioral Analytics Engine |
- | EA-0028 | IS-028 | AQELYN Cloud Security Posture Management Intelligence Engine |
- | EA-0029 | IS-029 | AQELYN SaaS Security Posture Management Intelligence Engine |

The fixed repository structure remains:

```
AQELYN/  
■■■■ archive/  
■■■■ blueprint/  
■■■■ docs/  
■■■■ src/  
■■■■ tests/  
■■■■ tools/  
■■■■ build/  
■■■■ releases/  
■■■■ scripts/  
■■■■ assets/  
■■■■ examples/  
■■■■ plugins/  
■■■■ sdk/  
■■■■ api/  
■■■■ README.md
```

2. IS-029 Specification Identity

Specification ID: IS-029

Name: AQELYN SaaS Security Posture Management (SSPM) Intelligence Engine

Engineering Archive Target: EA-0029

Project: AQELYN

System Type: Cyber Security Operating Environment

Status: Complete

Predecessor: IS-028 - AQELYN Cloud Security Posture Management Intelligence Engine

3. Purpose

The AQELYN SaaS Security Posture Management (SSPM) Intelligence Engine provides continuous assessment of Software-as-a-Service platforms, identifying security misconfigurations, excessive permissions, application-to-application trust risks, compliance violations, exposed data, identity weaknesses, and third-party integration risks across enterprise SaaS ecosystems.

The engine establishes continuous SaaS posture intelligence by correlating application configurations, identity context, audit telemetry, collaboration metadata, API integrations, and governance policies using explainable AI and policy-driven analytics.

It answers:

```
Which SaaS applications are misconfigured?  
Which SaaS tenants violate enterprise policy?  
Which SaaS identities have excessive permissions?  
Which third-party integrations introduce risk?  
Which SaaS platforms expose sensitive information?  
Can every SaaS posture assessment be explained and audited?
```

4. Mission

The engine shall provide:

```
Continuous SaaS posture assessment  
SaaS misconfiguration detection  
SaaS compliance validation  
SaaS identity analysis  
Third-party integration intelligence  
SaaS risk scoring  
Mission-aware SaaS reporting  
Executive SaaS security summaries  
Continuous reassessment  
SaaS governance support
```

5. Scope

5.1 In Scope

```
Microsoft 365  
Google Workspace  
Salesforce  
ServiceNow  
Slack  
Atlassian Cloud  
GitHub Enterprise Cloud  
Zoom  
Okta  
Enterprise SaaS platforms
```

5.2 Out of Scope

```
Custom application development  
SaaS licensing optimization  
Billing management  
Financial analytics  
End-user productivity metrics
```

6. Dependencies

IS-029 depends on:

IS-001 AQELYN Kernel
 IS-002 Universal Object Model
 IS-003 AQELYN Event Bus
 IS-004 AQELYN Evidence Engine
 IS-005 AQELYN Knowledge Graph
 IS-006 AQELYN Trust Engine
 IS-007 AQELYN Mission Engine
 IS-008 AQELYN Workflow Engine
 IS-009 AQELYN Policy Engine
 IS-010 AQELYN Compliance & Governance Engine
 IS-011 Identity & Access Governance Engine
 IS-012 Asset & Configuration Governance Engine
 IS-013 Risk Intelligence Engine
 IS-014 Threat Intelligence Fusion Engine
 IS-015 Security Operations Engine
 IS-016 Digital Forensics Engine
 IS-017 Threat Detection & Analytics Engine
 IS-018 Automated Response & Orchestration Engine
 IS-019 Security Data Lake & Telemetry Platform
 IS-020 AI Decision Intelligence Engine
 IS-021 Predictive Analytics & Forecasting Engine
 IS-022 Executive Intelligence & Strategic Reporting Engine
 IS-023 Threat Exposure & Attack Surface Management Engine
 IS-024 Vulnerability Intelligence & Prioritization Engine
 IS-025 Cyber Asset Discovery & Inventory Intelligence Engine
 IS-026 Configuration Compliance & Drift Intelligence Engine
 IS-027 Identity Threat Detection & Behavioral Analytics Engine
 IS-028 Cloud Security Posture Management Intelligence Engine

7. High-Level Architecture

AQELYN SaaS Security Posture Management Engine
 ■
 ■■■ SaaS Discovery Engine
 ■■■ SaaS Configuration Assessment Engine
 ■■■ SaaS Identity Intelligence Engine
 ■■■ SaaS Integration Risk Engine
 ■■■ SaaS Compliance Engine
 ■■■ SaaS Risk Engine
 ■■■ Knowledge Graph Connector
 ■■■ Security Data Lake Connector
 ■■■ AI Decision Connector
 ■■■ Executive Reporting Connector
 ■■■ Event Publisher

8. Functional Requirements

FR-029-001 - SaaS Discovery

The engine shall continuously discover:

SaaS tenants
 Applications
 Users
 Groups
 Roles
 Connected integrations
 OAuth applications
 Administrative accounts

FR-029-002 - SaaS Configuration Assessment

Continuously evaluate:

Security settings
 Authentication policies
 MFA configuration
 External sharing
 Administrative controls

Retention policies
Audit settings
Application configuration

FR-029-003 - SaaS Risk Detection

Detect:

Weak authentication
Excessive privileges
Unauthorized integrations
Public sharing
Inactive administrators
Risky OAuth applications
Policy violations

FR-029-004 - Explainable SaaS Intelligence

Every SaaS assessment shall include:

Evidence references
Confidence indicators
Policy rationale
Application metadata
Historical context
Risk explanation

FR-029-005 - Governance

Support:

Application governance
Approval workflows
Policy validation
Auditability
Executive review

FR-029-006 - Event Publication

Publish standardized events:

saas.application.discovered
saas.configuration.assessed
saas.policy.violation
saas.identity.risk.updated
saas.integration.detected
saas.remediation.recommended

9. Non-Functional Requirements

The engine shall provide:

Continuous assessment
Enterprise scalability
Low-latency SaaS analysis
Explainability
Auditability
Repository stability
Backward compatibility

10. Core SaaS Assessment Lifecycle

```

SaaS Application Discovered
↓
Configuration Assessment
↓
Identity Analysis
↓
Risk Assessment
↓
Compliance Validation
↓
Policy Enforcement
↓
Continuous Monitoring

```

11. Internal Component Architecture

The AQELYN SaaS Security Posture Management Intelligence Engine is implemented as a modular SaaS security intelligence platform integrated with the AQELYN Kernel, Knowledge Graph, Security Data Lake, Identity Governance Engine, Cloud Security Posture Management Engine, Risk Intelligence Engine, AI Decision Intelligence Engine, Executive Intelligence Engine, and Security Operations Engine.

```

AQELYN SaaS Security Posture Management Engine
■
■■■ SaaS Discovery Engine
■■■ SaaS Configuration Assessment Engine
■■■ SaaS Identity Intelligence Engine
■■■ SaaS Integration Risk Engine
■■■ SaaS Compliance Engine
■■■ SaaS Risk Engine
■■■ Knowledge Graph Connector
■■■ Security Data Lake Connector
■■■ AI Decision Connector
■■■ Executive Reporting Connector
■■■ Event Publisher

```

12. Component Specifications

12.1 SaaS Discovery Engine

Continuously discovers enterprise SaaS environments.

Capabilities:

```

Tenant discovery
Application inventory
User enumeration
Role discovery
OAuth application discovery
Integration inventory

```

12.2 SaaS Configuration Assessment Engine

Continuously evaluates SaaS security configurations.

Supports:

```

Security policy validation
Authentication assessment
MFA verification
Sharing configuration analysis
Audit configuration validation
Configuration drift detection

```

12.3 SaaS Identity Intelligence Engine

Analyzes SaaS identities and permissions.

Produces:

```
Identity exposure
Privilege analysis
Role assessment
Permission inheritance
Identity risk scoring
```

12.4 SaaS Integration Risk Engine

Analyzes connected applications and integrations.

Produces:

```
OAuth application assessment
API trust analysis
Third-party integration risk
Application permissions
Integration anomaly detection
```

12.5 SaaS Compliance Engine

Validates SaaS security posture against governance requirements.

Supports:

```
Compliance scoring
Control validation
Regulatory mapping
Policy correlation
Audit evidence
```

12.6 SaaS Risk Engine

Calculates SaaS security risk.

Produces:

```
Application risk score
Mission impact
Business impact
Threat likelihood
Remediation priority
```

13. Universal Object Model Extensions

13.1 SaaSApplication

```
SaaSApplication:
  application_id
  vendor
  tenant
  status
```

13.2 SaaSAssessment

```
SaaSAssessment:
  assessment_id
  compliance_score
  risk_score
```

generated_at

13.3 SaaSIntegration

SaaSIntegration:
integration_id
provider
permission_scope
confidence

13.4 SaaSRemediation

SaaSRemediation:
remediation_id
recommendation
owner
target_date

14. Knowledge Graph Integration

Relationships:

SaaSApplication
↓
contains
↓
Identity

Identity
↓
uses
↓
Integration

Integration
↓
creates
↓
Risk

Risk
↓
requires
↓
Remediation

15. Event Bus Integration

15.1 SaaS Discovery Events

saas.application.discovered
saas.application.updated
saas.application.removed

15.2 Configuration Events

saas.configuration.assessed
saas.policy.violation

15.3 Identity Events

saas.identity.risk.updated
saas.permission.changed

15.4 Integration Events

```
saas.integration.detected  
saas.integration.risk.updated  
saas.remediation.recommended
```

16. Security Data Lake Integration

Consumes:

```
SaaS audit logs  
Authentication telemetry  
Administrative activity  
Application metadata  
API activity
```

17. AI Decision Intelligence Integration

Consumes:

```
Risk scores  
Integration intelligence  
Compliance trends  
Remediation recommendations  
Confidence indicators
```

18. Risk & Threat Integration

Consumes:

```
Threat intelligence  
Mission impact  
Business criticality  
Identity risk  
Cloud posture intelligence
```

19. Compliance Integration

Supports:

```
Application governance  
Regulatory mapping  
Control validation  
Audit evidence  
Executive reporting
```

20. Public APIs

20.1 SaaS Inventory API

```
GET /saas/applications  
GET /saas/applications/{id}
```

20.2 SaaS Assessment API

```
GET /saas/assessments
GET /saas/assessments/{id}
```

20.3 SaaS Risk API

```
GET /saas/risk
GET /saas/risk/{id}
```

20.4 SaaS Integration API

```
GET /saas/integrations
POST /saas/remediations
```

21. Repository Impact

Implementation shall use the approved repository structure.

```
AQELYN/
■■■ src/
■ ■■■ saas_security_posture/
■■■ tests/
■ ■■■ saas_security_posture/
■■■ docs/
■ ■■■ saas_security_posture/
■■■ api/
■ ■■■ saas_security_posture/
■■■ archive/
```

No top-level repository modifications are permitted.

22. Security Architecture

The AQELYN SaaS Security Posture Management Intelligence Engine is the trusted subsystem responsible for continuously assessing SaaS environments, validating security posture, detecting application misconfigurations, evaluating identity and integration risks, and governing enterprise SaaS security intelligence.

Every SaaS posture assessment shall be:

```
Explainable
Evidence-backed
Policy-governed
Application-aware
Risk-aware
Mission-aware
Fully auditable
Continuously reassessed
```

22.1 Security Principles

```
Zero Trust
Defense in Depth
Least Privilege
Continuous SaaS Assessment
Secure by Design
Policy Enforcement
Explainable SaaS Intelligence
Continuous Monitoring
```

22.2 Authorization Model

Supported operational roles:

```
SaaS Security Administrator
Application Administrator
Security Administrator
SOC Analyst
Compliance Officer
Mission Owner
Identity Administrator
Executive Reviewer
```

All SaaS posture assessments and remediation decisions shall be governed through the AQELYN Policy Engine.

22.3 SaaS Assessment Integrity

SaaS assessments shall maintain:

```
Unique assessment identifier
SaaS application reference
Tenant reference
Evidence references
Risk score
Compliance score
Confidence indicators
Remediation guidance
Audit trail
```

Assessment history shall be append-only.

22.4 Integration Integrity

SaaS integration records shall maintain:

```
Integration identifier
Provider
Permission scope
Trust relationship
Application owner
Risk state
Confidence metrics
Audit record
```

No SaaS integration shall be considered trusted without ownership, permission scope, and validation metadata.

23. SaaS Security Lifecycle

23.1 Assessment Lifecycle

```
SaaS Application Discovered
↓
Configuration Assessment
↓
Identity Analysis
↓
Integration Analysis
↓
Risk Assessment
↓
Continuous Monitoring
```

23.2 Remediation Lifecycle

```
Issue Detected
↓
Risk Prioritized
↓
Remediation Generated
↓
Approval Granted
```

```
↓  
Remediation Applied  
↓  
Validation Completed
```

23.3 Audit Lifecycle

```
Assessment Created  
↓  
Evidence Linked  
↓  
Policy Validated  
↓  
Audit Stored
```

24. Continuous SaaS Operations

The engine continuously evaluates:

```
SaaS applications  
Tenant configurations  
Identity posture  
Administrative accounts  
OAuth applications  
API integrations  
Compliance posture  
Application exposure
```

25. Performance Requirements

The engine shall support:

```
Continuous assessment  
Low-latency SaaS analysis  
Enterprise-scale SaaS environments  
Concurrent application assessments  
High availability  
Continuous operation
```

26. Scalability Requirements

The engine shall scale to support:

```
Global enterprises  
Thousands of SaaS applications  
Millions of user identities  
Distributed SaaS assessments  
Long-term posture history  
Hybrid SaaS ecosystems
```

27. Audit Requirements

Every SaaS posture operation shall generate immutable audit records.

Audit events include:

```
Application discovered  
Configuration assessed
```

Identity risk updated
Integration detected
Policy violation identified
Remediation completed

28. Failure Handling

28.1 Assessment Failure

Assessment retried
Failure recorded
Administrator notified

28.2 Compliance Failure

Validation retried
Previous assessment retained
Audit generated

28.3 Integration Analysis Failure

Analysis repeated
Manual review initiated
Audit recorded

28.4 Policy Failure

Application approval blocked
Policy violation recorded
Manual approval required

29. Testing Strategy

29.1 Unit Testing

Validate:

SaaS Discovery Engine
SaaS Configuration Assessment Engine
SaaS Identity Intelligence Engine
SaaS Integration Risk Engine
SaaS Compliance Engine
SaaS Risk Engine

29.2 Integration Testing

Verify interaction with:

Kernel
Knowledge Graph
Security Data Lake
Identity Governance Engine
Cloud Security Posture Management Engine
Risk Intelligence Engine
AI Decision Engine
Executive Reporting Engine
Security Operations Engine

29.3 System Testing

Validate:

```
Application discovery
Configuration assessment
Identity analysis
Integration analysis
Risk scoring
Audit generation
```

29.4 Security Testing

Verify:

```
Authorization
Policy enforcement
Explainability
Audit logging
SaaS assessment integrity
```

29.5 Regression Testing

Verify IS-001 through IS-028 remain unaffected.

30. Acceptance Criteria

IS-029 is complete when:

```
SaaS Discovery Engine implemented
SaaS Configuration Assessment Engine implemented
SaaS Identity Intelligence Engine implemented
SaaS Integration Risk Engine implemented
SaaS Compliance Engine implemented
SaaS Risk Engine implemented
Repository unchanged
Testing documented
```

31. Repository Validation

Repository structure remains unchanged.

```
AQELYN/
■■■ src/saas_security_posture/
■■■ tests/saas_security_posture/
■■■ docs/saas_security_posture/
■■■ api/saas_security_posture/
■■■ archive/
```

No top-level repository modifications are permitted.

32. Engineering Summary

IS-029 introduces the AQELYN SaaS Security Posture Management Intelligence Engine, providing enterprise-scale SaaS security posture assessment, identity analysis, third-party integration intelligence, compliance validation, explainable SaaS risk scoring, and continuous SaaS governance.

Major capabilities include:

```
SaaS Application Discovery
SaaS Configuration Assessment
```

SaaS Identity Analysis
Third-Party Integration Intelligence
SaaS Compliance Validation
SaaS Risk Scoring
Mission-Aware SaaS Reporting
Executive SaaS Security Reporting
Continuous SaaS Monitoring
SaaS Governance

The engine integrates with all previously completed AQELYN engines while preserving repository stability, modularity, and backward compatibility.

33. Specification Status

Specification ID : IS-029
Title : AQELYN SaaS Security Posture Management (SSPM) Intelligence Engine
Status : COMPLETE
Engineering Archive : READY FOR GENERATION
Next Artifact : EA-0029

34. EA-0029 Engineering Objective

The objective of IS-029 was to introduce a dedicated SaaS Security Posture Management Intelligence Engine that enables AQELYN to continuously discover SaaS applications, assess tenant configuration, analyze SaaS identity posture, evaluate third-party integrations, validate SaaS compliance, score SaaS risk, and generate remediation recommendations.

The engine extends AQELYN from cloud security posture into SaaS security posture and enterprise application governance.

35. EA-0029 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

SaaS discovery
SaaS configuration assessment
SaaS identity intelligence
SaaS integration risk analysis
SaaS compliance validation
SaaS risk scoring
SaaS remediation recommendations
Knowledge Graph integration
Security Data Lake integration
Risk and threat intelligence integration
AI Decision integration
Executive reporting integration
Event publishing

36. Major Engineering Decisions

36.1 Decision 1 - Dedicated SaaS Security Posture Engine

SaaS security posture responsibilities are implemented as a standalone engine rather than embedded in cloud posture, identity governance, or security operations.

Rationale:

Clear separation of SaaS tenant posture from cloud infrastructure posture.
 Independent lifecycle and governance.
 Better support for application configuration, identity, and integration risks.
 Improved traceability for SaaS posture assessments.

36.2 Decision 2 - Integrations Are First-Class Risk Objects

Third-party and OAuth integrations are modeled as risk-bearing entities.

Benefits:

Application-to-application trust becomes visible.
 Permission scopes can be assessed and audited.
 Risky integrations can be prioritized for review.

36.3 Decision 3 - SaaS Assessments Are Evidence-Backed

Every SaaS assessment includes evidence, policy rationale, application metadata, historical context, confidence indicators, and risk explanation.

Benefits:

SaaS posture decisions become defensible.
 Security teams can explain findings.
 Compliance teams can audit SaaS governance decisions.

36.4 Decision 4 - Event-Driven SaaS Posture Lifecycle

SaaS discovery, configuration, identity, integration, risk, and remediation events are published through the AQELYN Event Bus.

Examples include:

```
saas.application.discovered
saas.application.updated
saas.application.removed
saas.configuration.assessed
saas.policy.violation
saas.identity.risk.updated
saas.permission.changed
saas.integration.detected
saas.integration.risk.updated
saas.remediation.recommended
```

36.5 Decision 5 - Universal Object Model Extension

New domain objects introduced include:

```
SaaSApplication
SaaSAssessment
SaaSIntegration
SaaSRemediation
```

37. Architectural Integration Summary

| Engine | Integration |

|---|---|

| IS-001 Kernel | Runtime lifecycle and service registration |

| IS-002 Universal Object Model | SaaS application, assessment, integration, remediation objects |

| IS-003 Event Bus | SaaS discovery, configuration, identity, integration, remediation events |

IS-004 Evidence Engine	Evidence references and SaaS support
IS-005 Knowledge Graph	SaaS, identity, integration, risk, remediation relationships
IS-006 Trust Engine	Data confidence and SaaS assessment trust
IS-007 Mission Engine	Mission-aware SaaS risk prioritization
IS-008 Workflow Engine	SaaS remediation and approval workflows
IS-009 Policy Engine	SaaS governance and policy validation
IS-010 Compliance Engine	SaaS compliance and regulatory mapping
IS-011 Identity Governance Engine	SaaS identities and roles
IS-013 Risk Intelligence Engine	Risk scoring and business impact
IS-014 Threat Intelligence Engine	SaaS threat and integration context
IS-015 Security Operations Engine	SaaS investigations and SOC workflows
IS-019 Security Data Lake	SaaS audit logs and API telemetry
IS-020 AI Decision Engine	Remediation recommendations and confidence scoring
IS-022 Executive Reporting Engine	Executive SaaS summaries
IS-028 Cloud Security Posture Engine	Cloud posture and identity context

No existing engine required redesign.

38. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```
AQELYN/  
■■■ src/saas_security_posture/  
■■■ tests/saas_security_posture/  
■■■ api/saas_security_posture/  
■■■ docs/saas_security_posture/  
■■■ archive/
```

No top-level directories were added, removed, or renamed.

39. Security Impact Summary

The specification introduces SaaS-posture-specific security controls:

```
Policy-governed SaaS assessments  
Evidence-backed application findings  
Continuous SaaS monitoring  
SaaS identity risk analysis  
Third-party integration risk analysis  
SaaS compliance validation  
Remediation ownership tracking  
SaaS assessment audit trail  
Role-authorized SaaS posture administration
```

No reduction in the security posture of existing components was identified.

40. Capabilities Added

The engine enables AQELYN to support:

```
Continuous SaaS posture assessment
SaaS misconfiguration detection
SaaS compliance validation
SaaS identity analysis
Third-party integration intelligence
SaaS risk scoring
Mission-aware SaaS reporting
Executive SaaS security summaries
Continuous reassessment
SaaS governance support
```

41. Risks Identified

| Risk | Mitigation |

|---|---|

| Unapproved SaaS integrations | Integration inventory and risk scoring |

| Excessive SaaS permissions | SaaS Identity Intelligence Engine |

| Public sharing exposure | Configuration Assessment Engine |

| Incomplete SaaS visibility | SaaS discovery and audit telemetry ingestion |

| Weak authentication posture | Authentication and MFA assessment |

| Stale application posture | Continuous reassessment |

| Unauthorized remediation | Policy enforcement and workflow approval |

| Poor auditability | Evidence-backed assessments and audit events |

No critical architectural risks were identified that require redesign.

42. Verification Summary

The specification defines verification for:

```
Unit testing
Integration testing
System testing
Security testing
Regression testing
```

Acceptance criteria cover SaaS discovery, configuration assessment, identity intelligence, integration risk, compliance, SaaS risk, repository validation, and testing documentation.

43. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

```
Modular architecture
Event-driven communication
Immutable evidence references
Traceability
Explainability
Security by design
Repository stability
Backward compatibility
Governance-before-archive discipline
```

44. Dependencies

Required:

EA-0001 through EA-0028
IS-001 through IS-029

Enables:

IS-030 and subsequent software supply chain, SBOM intelligence, and cyber resilience components

45. Completion Record

Engineering Archive : EA-0029
Implementation Specification : IS-029
Title : AQELYN SaaS Security Posture Management Intelligence Engine
Engineering Status : COMPLETE
Repository Status : UNCHANGED
Architecture Status : EXTENDED
Backward Compatibility : MAINTAINED

46. Archive Index Update

EA-0001 IS-001 AQELYN Kernel
EA-0002 IS-002 Universal Object Model
EA-0003 IS-003 AQELYN Event Bus
EA-0004 IS-004 AQELYN Evidence Engine
EA-0005 IS-005 AQELYN Knowledge Graph
EA-0006 IS-006 AQELYN Trust Engine
EA-0007 IS-007 AQELYN Mission Engine
EA-0008 IS-008 AQELYN Workflow Engine
EA-0009 IS-009 AQELYN Policy Engine
EA-0010 IS-010 AQELYN Compliance & Governance Engine
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EA-0026 IS-026 AQELYN Configuration Compliance & Drift Intelligence Engine
EA-0027 IS-027 AQELYN Identity Threat Detection & Behavioral Analytics Engine
EA-0028 IS-028 AQELYN Cloud Security Posture Management Intelligence Engine
EA-0029 IS-029 AQELYN SaaS Security Posture Management Intelligence Engine

47. Engineering Phase Status

Completed Engineering Archives : EA-0001 through EA-0029

Current Status:
EA-0029 COMPLETE

Next Implementation Specification:
IS-030 - AQELYN Software Supply Chain Security & SBOM Intelligence Engine

48. Engineering Archive Publication Standard

EA-0029 follows the AQELYN Engineering Archive Publication Standard.

Required package structure:

```
EA-xxxx/  
■  
■ ■ ■ ■ diagrams/  
■ ■ ■ ■ Architecture.svg  
■ ■ ■ ■ Component.svg  
■ ■ ■ ■ Workflow.svg  
■ ■ ■ ■ EventFlow.svg  
■ ■ ■ ■ Integration.svg  
■  
■ ■ ■ ■ examples/  
■ ■ ■ ■ example_*.md  
■  
■ ■ ■ ■ HTML/  
■ ■ ■ ■ index.html  
■ ■ ■ ■ styles.css  
■ ■ ■ ■ assets/  
■  
■ ■ ■ ■ images/  
■  
■ ■ ■ ■ journal/  
■ ■ ■ ■ Engineering_Journal.md  
■  
■ ■ ■ ■ manifest/  
■ ■ ■ ■ manifest.json  
■  
■ ■ ■ ■ MD/  
■ ■ ■ ■ EA-xxxx.md  
■  
■ ■ ■ ■ PDF/  
■ ■ ■ ■ EA-xxxx.pdf  
■  
■ ■ ■ ■ requirements/  
■ ■ ■ ■ Requirements_Matrix.md  
■  
■ ■ ■ ■ traceability/  
■ ■ ■ ■ Traceability_Matrix.md  
■  
■ ■ ■ ■ README.md
```

The master Markdown is the source of truth. The PDF and HTML are generated from the same master Markdown and must not omit sections.

49. Requirements Matrix

Requirement ID Requirement Evidence in Archive Status			
--- --- --- ---			
FR-029-001	Continuously discover SaaS applications	Sections 8, 12, 23	Complete
FR-029-002	Assess SaaS configuration	Sections 8, 12, 23	Complete
FR-029-003	Detect SaaS risks	Sections 8, 12, 23	Complete
FR-029-004	Provide explainable SaaS intelligence	Sections 8, 22, 36	Complete
FR-029-005	Support governance	Sections 8, 22, 23	Complete
FR-029-006	Publish SaaS events	Sections 8, 15, 36	Complete
NFR-029-001	Continuous assessment	Sections 9, 24	Complete
NFR-029-002	Enterprise scalability	Sections 9, 25, 26	Complete
NFR-029-003	Low-latency SaaS analysis	Sections 9, 25	Complete
NFR-029-004	Explainability	Sections 9, 22, 36	Complete
NFR-029-005	Auditability	Sections 9, 27, 39	Complete
NFR-029-006	Repository stability	Sections 21, 31, 38	Complete

50. Traceability Matrix

| Source | Target | Relationship |

|---|---|---|

| IS-029 Purpose | EA-0029 Objective | Defines why the engine exists |

| SaaS Discovery Engine | FR-029-001 | Discovers SaaS applications and tenants |

| SaaS Configuration Assessment Engine | FR-029-002 | Assesses application settings |

| SaaS Identity Intelligence Engine | FR-029-003 | Analyzes identities and permissions |

| SaaS Integration Risk Engine | FR-029-003 | Evaluates third-party integrations |

| SaaS Compliance Engine | Compliance validation | Maps SaaS posture to controls |

| SaaS Risk Engine | Risk scoring | Calculates SaaS security risk |

| Event Publisher | FR-029-006 | Publishes SaaS events |

| Security Data Lake Integration | SaaS telemetry | Supplies audit and API data |

| AI Decision Integration | Remediation recommendations | Supplies confidence and recommendations |

| Risk & Threat Integration | Threat context | Supplies threat and business context |

| Compliance Integration | Governance and audit | Supports regulatory reporting |

| Repository Validation | Repository Standard | Confirms no top-level redesign |

51. Engineering Journal

Journal Entry - EA-0029

EA-0029 was created to archive completion of IS-029 - AQELYN SaaS Security Posture Management Intelligence Engine.

The archive records the expansion of AQELYN into SaaS posture management. IS-029 defines the structure needed to discover SaaS applications and tenants, assess configurations, analyze identities, evaluate third-party integrations, validate compliance, calculate SaaS risk, generate remediation recommendations, and publish SaaS posture events.

The engineering design preserves the fixed AQELYN repository structure and maintains backward compatibility with previously completed engines.

Lessons Learned

SaaS posture must be modeled separately from cloud posture. Cloud posture assesses infrastructure and provider resources; SaaS posture assesses application tenants, users, permissions, external sharing, integrations, and application governance.

Governance Note

EA-0029 follows the master-document publication workflow. The Markdown file is the authoritative source, and PDF/HTML representations are generated from the same content.

52. Examples

52.1 Example SaaS Application

```
application_id: SAAS-0001
vendor: Microsoft 365
tenant: enterprise-primary
```

```
status: active
owner: collaboration_security_team
```

52.2 Example SaaS Assessment

```
assessment_id: SAAS-ASSESS-2001
compliance_score: 84
risk_score: 72
generated_at: 2026-07-07T12:00:00Z
```

52.3 Example SaaS Integration

```
integration_id: INT-3001
provider: third_party_oauth_app
permission_scope: read_write_mailbox
confidence: 0.89
risk_state: high
```

52.4 Example SaaS Event

```
{
  "event_type": "saas.policy.violation",
  "application_id": "SAAS-0001",
  "risk_score": 72,
  "source_engine": "aqelyn_saas_security_posture_management_engine"
}
```

53. Manifest Summary

Archive contents include:

```
README.md
MD/EA-0029.md
PDF/EA-0029.pdf
HTML/index.html
HTML/styles.css
manifest/manifest.json
requirements/Requirements_Matrix.md
traceability/Traceability_Matrix.md
journal/Engineering_Journal.md
diagrams/Architecture.svg
diagrams/Component.svg
diagrams/Workflow.svg
diagrams/EventFlow.svg
diagrams/Integration.svg
examples/example_saas_security_posture.md
```

54. Final Archive Statement

EA-0029 is the Engineering Archive for IS-029 - AQELYN SaaS Security Posture Management Intelligence Engine.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the engineering publication standard.

```
EA-0029 COMPLETE
IS-029 COMPLETE
NEXT: IS-030
```